

Exercise 1

1. Write a JDBC application which will interact with Database and perform the following task.

- a. Create Student Table with Roll No, Name, and Address field and insert few records.
- b. Using Statement Object display the content of Record.
- c. Using Statement Object Insert Two Record.
- d. Using Statement Object Update One Record.
- e. Using Statement Object Delete One Record.
- f. Using Statement Object display the content of Record.

a). Create Table & Insert Few Records

//Creation of Database

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.Statement;

public class CreateDatabase {
    public static void main(String[] args) {
        // SQL command to create a database
        String sql = "CREATE DATABASE COLLEGE";
        try {
            //Step 1: Load the Driver
            Class.forName("com.mysql.cj.jdbc.Driver");
            // Step 2: Connect to MySQL server
            Connection conn = DriverManager.getConnection("jdbc:mysql://localhost:3306/", "root", "root");
            // Step 3: Create a Statement object
            Statement stmt = conn.createStatement();

            // Step 4: Execute the SQL statement
            stmt.executeUpdate(sql);
            System.out.println("Database created successfully...");

            // Step 5: Close connection
            stmt.close();
            conn.close();
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

// Create Student Table with Roll No, Name, and Address field and insert few records

```
import java.sql.*;
public class Program1a
{
    public static void main(String[] args) throws ClassNotFoundException
    {
        try
        {
            // Step 1: Load and register JDBC driver (optional in newer versions)
            Class.forName("com.mysql.cj.jdbc.Driver");
            // Step 2: Connect to MySQL server
            Connection conn = DriverManager.getConnection("jdbc:mysql://localhost:3306/college", "root",
"root");
            // Step 3: Create a Statement object
            Statement stmt = conn.createStatement();
            // Step 4: Execute the SQL statement
            //Create table student and execute
            String createTableSQL = "CREATE TABLE Student( Rollno INT, Name VARCHAR(50), Address
VARCHAR(50))";
            stmt.executeUpdate(createTableSQL);
            //inserting fields into table
            String insert = "INSERT INTO Student VALUES (1, 'Ramesh', 'Delhi')," + "(2, 'Bobby',
'Mumbai')";
            stmt.executeUpdate(insert);
            System.out.println("Table created and records inserted.");
            // Step 5: Close connection
            stmt.close();
            conn.close();
        }
        catch (SQLException e)
        {
            e.printStackTrace();
        }
    }
}
```

b)Using Statement Object display the content of Record.

//Using Statement Object display the content of Record.

```
import java.sql.*;
public class Program1b
{
    public static void main(String[] args) throws ClassNotFoundException
    {
        try
        {
            // Step 1: Load and register JDBC driver
            Class.forName("com.mysql.cj.jdbc.Driver");
            // Step 2: Connect to MySQL Server 'College' Database
            Connection conn = DriverManager.getConnection("jdbc:mysql://localhost:3306/college", "root",
"root");
            // Step 3: Create a Statement object
            Statement stmt = conn.createStatement();
            // Step 4: Execute the SQL statement
            ResultSet rs = stmt.executeQuery("SELECT * FROM student");
            //displaying result set data
            while (rs.next())
            {
                int roll = rs.getInt("rollno");
                String name = rs.getString("name");
                String address = rs.getString("address");
                System.out.println(roll + " | " + name + " | " + address);
            }
            // Step 5: Close connection
            conn.close();
        }
        catch (SQLException e)
        {
            e.printStackTrace();
        }
    }
}
```

c)Using Statement Object Insert Two Record.

```
import java.sql.*;

public class Program1c
{
    public static void main(String[] args) throws ClassNotFoundException
    {
        try
        {
            // Step 1: Load and register JDBC driver (optional in newer versions)
            Class.forName("com.mysql.cj.jdbc.Driver");

            // Step 2: Connect to MySQL server
            Connection conn = DriverManager.getConnection("jdbc:mysql://localhost:3306/college", "root",
"root");

            // Step 3: Create a Statement object
            Statement stmt = conn.createStatement();

            ResultSet rs = stmt.executeQuery("SELECT * FROM student");
            //displaying result set data before inserting fields into table
            while (rs.next())
            {
                int roll = rs.getInt("rollno");
                String name = rs.getString("name");
                String address = rs.getString("address");
                System.out.println(roll + " | " + name + " | " + address);
            }

            //inserting two records

            String insert1 = "INSERT INTO student VALUES (3, 'Charlie', 'Chennai')";
            String insert2 = "INSERT INTO student VALUES (4, 'Diana', 'Bangalore')";

            // Step 4: Execute the SQL statement
            stmt.executeUpdate(insert1);
            stmt.executeUpdate(insert2);

            System.out.println("Two records inserted.");

            ResultSet rs1 = stmt.executeQuery("SELECT * FROM student");
            //displaying result set data after two records inserted into table
            while (rs1.next())
            {
                int roll = rs1.getInt("rollno");
                String name = rs1.getString("name");
```

```
        String address = rs1.getString("address");
        System.out.println(roll + " | " + name + " | " + address);
    }
    // Step 5: Close connection
    conn.close();
}
catch (SQLException e)
{
    e.printStackTrace();
}
}
}
```

d). Using Statement Object Update One Record and Displaying the records

```
import java.sql.*;

public class Program1b
{
    public static void main(String[] args) throws ClassNotFoundException
    {
        try
        {
            // Step 1: Load and register JDBC driver
            Class.forName("com.mysql.cj.jdbc.Driver");

            // Step 2: Connect to MySQL Server 'College' Database
            Connection conn = DriverManager.getConnection("jdbc:mysql://localhost:3306/college", "root",
"root");

            // Step 3: Create a Statement object
            Statement stmt = conn.createStatement();
            ResultSet rs = stmt.executeQuery("SELECT * FROM student");
            //displaying result set data
            while (rs.next())
            {
                int roll = rs.getInt("rollno");
                String name = rs.getString("name");
                String address = rs.getString("address");
                System.out.println(roll + " | " + name + " | " + address);
            }

            // Step 4: Execute the SQL statement
            String update = "UPDATE student SET address = 'Hyderabad' WHERE rollno = 2";
            stmt.executeUpdate(update);
            ResultSet rs1 = stmt.executeQuery("SELECT * FROM student");
            //displaying result set data
            System.out.println("After Updating:\n");
            while (rs1.next())
            {
                int roll = rs1.getInt("rollno");
                String name = rs1.getString("name");
                String address = rs1.getString("address");
                System.out.println(roll + " | " + name + " | " + address);
            }
        }
    }
}
```

```
// Step 5: Close connection
    conn.close();
}
catch (SQLException e)
{
    e.printStackTrace();
}
}
```

e) Using Statement Object Delete One Record and Displaying the contents of records

```
import java.sql.*;

public class Program1e
{
    public static void main(String[] args) throws ClassNotFoundException
    {
        try
        {
            // Step 1: Load and register JDBC driver
            Class.forName("com.mysql.cj.jdbc.Driver");

            // Step 2: Connect to MySQL Server 'College' Database
            Connection conn = DriverManager.getConnection("jdbc:mysql://localhost:3306/college", "root",
"root");

            // Step 3: Create a Statement object
            Statement stmt = conn.createStatement();

            ResultSet rs = stmt.executeQuery("SELECT * FROM student");
            //displaying result set data Before delete
            while (rs.next())
            {
                int roll = rs.getInt("rollno");
                String name = rs.getString("name");
                String address = rs.getString("address");
                System.out.println(roll + " | " + name + " | " + address);
            }

            // Step 4: Execute the SQL statement
            String delete = "DELETE FROM student WHERE rollno = 1";
            stmt.executeUpdate(delete);
            System.out.println("after deleting record");
            ResultSet rs1 = stmt.executeQuery("SELECT * FROM student");
            //displaying result set data after delete
            while (rs1.next())
            {
                int roll = rs1.getInt("rollno");
                String name = rs1.getString("name");
                String address = rs1.getString("address");
                System.out.println(roll + " | " + name + " | " + address);
            }
        }
    }
}
```

```
    }  
    // Step 5: Close connection  
    conn.close();  
    }  
catch (SQLException e)  
{  
    e.printStackTrace();  
}  
}  
}
```

f. Using Statement Object display the content of Record.

```
import java.sql.*;
public class lab1f
{
    public static void main(String[] args) throws ClassNotFoundException
    {
        try
        {
            // Step 1: Load and register JDBC driver
            Class.forName("com.mysql.cj.jdbc.Driver");
            // Step 2: Connect to MySQL Server 'College' Database
            Connection conn = DriverManager.getConnection("jdbc:mysql://localhost:3306/college", "root",
"root");
            // Step 3: Create a Statement object
            Statement stmt = conn.createStatement();
            // Step 4: Execute the SQL statement
            ResultSet rs = stmt.executeQuery("SELECT * FROM student");
            //displaying result set data
            while (rs.next())
            {
                int roll = rs.getInt("rollno");
                String name = rs.getString("name");
                String address = rs.getString("address");
                System.out.println(roll + " | " + name + " | " + address);
            }
            // Step 5: Close connection
            conn.close();
        }
        catch (SQLException e)
        {
            e.printStackTrace();
        }
    }
}
```

Exercise 2

2. Write a JDBC application which will interact with Database and perform the following task.

1. Create Student Table with Roll No, Name, and Address field and insert few records.
2. Using Prepared Statement Object display the content of Record.
3. Using Prepared Statement Object Insert Two Record.
4. Using Prepared Statement Object Update One Record.
5. Using Prepared Statement Object Delete One Record.
6. Using Prepared Statement Object display the content of Record

//a. Create Student Table with Roll No, Name, and Address field and insert few records.

```
import java.sql.*;
public class Lab2a
{
    public static void main(String[] args) throws ClassNotFoundException
    {
        try
        {
            // Step 1: Load and register JDBC driver
            Class.forName("com.mysql.cj.jdbc.Driver");
            // Step 2: Connect to MySQL Server 'College' Database
            Connection conn = DriverManager.getConnection("jdbc:mysql://localhost:3306/college",
"root", "root");
            //Step3:Create Table StudentData using prepared Statement
            String createSQL = "CREATE TABLE IF NOT EXISTS StudentData(" + "RollNo INT
PRIMARY KEY, " + "Name VARCHAR(100), " + "Address VARCHAR(255))";
            PreparedStatement stmt = conn.prepareStatement(createSQL);
            stmt.executeUpdate();
            System.out.println("StudentData table created.");
            //Step4:Inserting values into StudentData table
            String insertSQL = "INSERT INTO StudentData(RollNo, Name, Address) VALUES (?, ?, ?)";
            PreparedStatement pstmt = conn.prepareStatement(insertSQL);
            pstmt.setInt(1, 1);
            pstmt.setString(2, "Alice Johnson");
            pstmt.setString(3, "123 Main St");
            pstmt.executeUpdate();
            pstmt.setInt(1, 2);
            pstmt.setString(2, "Bob Smith");
            pstmt.setString(3, "456 Oak St");
            pstmt.executeUpdate();
            System.out.println("Initial records inserted.");
            //Step5:Close the connection
            conn.close();
        }
        catch (SQLException e)
        {
            e.printStackTrace();
        }
    }
}
```

//b. Using Prepared Statement Object display the content of Record.

```

import java.sql.*;
public class Lab2b
{
    public static void main(String[] args) throws ClassNotFoundException
    {
        try
        {
            // Step 1: Load and register JDBC driver
            Class.forName("com.mysql.cj.jdbc.Driver");
            // Step 2: Connect to MySQL Server 'College' Database
            Connection conn = DriverManager.getConnection("jdbc:mysql://localhost:3306/college", "root",
"root");
            // Step 3: Create a Statement object
            String selectSQL = "SELECT * FROM StudentData";
            PreparedStatement stmt = conn.prepareStatement(selectSQL);
            // Step 4: Execute the SQL statement
            ResultSet rs = stmt.executeQuery();
            //displaying result set data
            // Print headers
            System.out.printf("%-8s %-15s %-20s\n", "RollNo", "Name", "Address");
            // Print each record
            while (rs.next())
            {
                System.out.printf("%-8d %-15s %-
20s\n",rs.getInt("RollNo"),rs.getString("Name"),rs.getString("Address"));
            }
            // Step 5: Close connection
            conn.close();
        }

        catch (SQLException e)
        {
            e.printStackTrace();
        }
    }
}

```

//c. Using Prepared Statement Object Insert Two Record.

```
import java.sql.*;

public class Lab2c
{
    public static void main(String[] args) throws ClassNotFoundException
    {
        try
        {
            // Step 1: Load and register JDBC driver
            Class.forName("com.mysql.cj.jdbc.Driver");
            // Step 2: Connect to MySQL Server 'College' Database
            Connection conn = DriverManager.getConnection("jdbc:mysql://localhost:3306/college", "root",
"root");
            String insertSQL = "INSERT INTO StudentData(RollNo, Name, Address) VALUES (?, ?, ?)";
            //Step3: Create Statement
            PreparedStatement stmt = conn.prepareStatement(insertSQL);
            //step4: Insert two records into table
            stmt.setInt(1, 3);
            stmt.setString(2, "Charlie Brown");
            stmt.setString(3, "789 Pine St");
            stmt.executeUpdate();

            stmt.setInt(1, 4);
            stmt.setString(2, "Diana Prince");
            stmt.setString(3, "321 Maple St");
            stmt.executeUpdate();

            System.out.println("Two new records inserted.");
            //Step5:Close Connection
            conn.close();
        }
        catch (SQLException e)
        {
            e.printStackTrace();
        }
    }
}
```

//d. Using Prepared Statement Object Update One Record.

```
import java.sql.*;

public class Lab2d
{
    public static void main(String[] args) throws ClassNotFoundException
    {
        try
        {
            // Step 1: Load and register JDBC driver
            Class.forName("com.mysql.cj.jdbc.Driver");
            // Step 2: Connect to MySQL Server 'College' Database
            Connection conn = DriverManager.getConnection("jdbc:mysql://localhost:3306/college", "root",
"root");
            String updateSQL = "UPDATE StudentData SET Address = ? WHERE RollNo = ?";
            //Step3:Create Statement
            PreparedStatement stmt = conn.prepareStatement(updateSQL);
            //Step4:Executing queries
            stmt.setString(1, "999 Updated Address");
            stmt.setInt(2, 2); // Update second record
            stmt.executeUpdate();
            System.out.println("Record updated.");
            //Step5:Close Connection
            conn.close();
        }
        catch (SQLException e) {
            e.printStackTrace();
        }
    }
}
```

//e. Using Prepared Statement Object Delete One Record.

```
import java.sql.*;
public class Lab2e {
    public static void main(String[] args)throws ClassNotFoundException
    {
        try
        {
            // Step 1: Load and register JDBC driver
            Class.forName("com.mysql.cj.jdbc.Driver");
            // Step 2: Connect to MySQL Server 'College' Database
            Connection conn = DriverManager.getConnection("jdbc:mysql://localhost:3306/college", "root",
"root");
            //Step3:Create Statement
            String deleteSQL = "DELETE FROM StudentData WHERE RollNo = ?";
            PreparedStatement stmt = conn.prepareStatement(deleteSQL);
            //Step4:Executing queries
            stmt.setInt(1, 1); // Delete Alice
            stmt.executeUpdate();
            System.out.println("Record deleted.");
            //Step5:Close Connection
            conn.close();
        }
        catch (SQLException e)
        {
            e.printStackTrace();
        }
    }
}
```

f).Using Prepared Statement Object display the content of Record

```
import java.sql.*;
public class Lab2f
{
    public static void main(String[] args) throws ClassNotFoundException
    {
        try
        {
            // Step 1: Load and register JDBC driver
            Class.forName("com.mysql.cj.jdbc.Driver");
            // Step 2: Connect to MySQL Server 'College' Database
            Connection conn = DriverManager.getConnection("jdbc:mysql://localhost:3306/college", "root",
"root");
            // Step 3: Create a Statement object
            String selectSQL = "SELECT * FROM StudentData";
            PreparedStatement stmt = conn.prepareStatement(selectSQL);
            // Step 4: Execute the SQL statement
            ResultSet rs = stmt.executeQuery();
            //displaying result set data
            // Print headers
            System.out.printf("%-8s %-15s %-20s\n", "RollNo", "Name", "Address");
            // Print each record
            while (rs.next())
            {
                System.out.printf("%-8d %-15s %-20s\n",rs.getInt("RollNo"),rs.getString("Name"),rs.getString("Address"));
            }
            // Step 5: Close connection
            conn.close();
        }

        catch (SQLException e)
        {
            e.printStackTrace();
        }
    }
}
```

Exercise 3**3. Write a JDBC application which will interact with Database and perform the following task.**

1. Create a store procedure which will insert one record into employee table.
2. Create a store procedure which will retrieve salary for given employee id.
3. Write a java application which will call the above procedure and display appropriate information on screen

Step1:**//Creating Employee Table**

```
import java.sql.*;

public class CreateEmpTable {
    public static void main(String[] args)throws ClassNotFoundException
    {
        try
        {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection conn = DriverManager.getConnection("jdbc:mysql://localhost:3306/college", "root",
"root");
            String sql = "CREATE TABLE IF NOT EXISTS employee (" + "emp_id INT PRIMARY KEY, "+
"name VARCHAR(100), "+ "salary DECIMAL(10, 2))";
            Statement stmt = conn.createStatement();
            stmt.execute(sql);
            System.out.println("Table created successfully.");

        }
        catch (SQLException e)
        {
            System.out.println("Error: " + e.getMessage());
        }
    }
}
```

//a) Procedure to insert one employee record

//b) Procedure to get salary by employee ID

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;
import java.sql.Statement;
public class ProcedureManager
{
    public static void createProcedures(Connection conn) throws SQLException,
    ClassNotFoundException
    {
        Class.forName("com.mysql.cj.jdbc.Driver");
        Statement stmt = conn.createStatement();
        // a. create insert_employee procedure
        stmt.execute("DROP PROCEDURE IF EXISTS insert_employee");
        stmt.execute("CREATE PROCEDURE insert_employee(IN id INT, IN name VARCHAR(100), IN
        sal DECIMAL(10,2)) " + "BEGIN INSERT INTO employee VALUES (id, name, sal); END");
        // b. create get_salary_by_emp_id procedure
        stmt.execute("DROP PROCEDURE IF EXISTS get_salary_by_emp_id");
        stmt.execute("CREATE PROCEDURE get_salary_by_emp_id(IN id INT, OUT sal
        DECIMAL(10,2)) " + "BEGIN SELECT salary INTO sal FROM employee WHERE emp_id = id;
        END");
        System.out.println(" Procedures created successfully.");
    }
}
```

c. Write a java application which will call the above procedure and display appropriate information on screen

//Step 3: Java Application to call these stored procedures

```
import java.sql.*;
import java.util.Scanner;

public class EmployeeApp
{
    public static void main(String[] args)
    {
        try
        {
            Connection conn = DriverManager.getConnection("jdbc:mysql://localhost:3306/college","root",
            "root");
            Scanner sc = new Scanner(System.in); // ✓ Scanner declared properly here
            // Call procedure creation from separate file
            ProcedureManager.createProcedures(conn);

            // Get input
            System.out.print("Enter ID: ");
            int id = sc.nextInt();
            sc.nextLine(); // clear buffer
            System.out.print("Enter Name: ");
            String name = sc.nextLine();
            System.out.print("Enter Salary: ");
            double salary = sc.nextDouble();

            // Insert employee
            CallableStatement insert = conn.prepareCall("{CALL insert_employee(?, ?, ?)}");
            insert.setInt(1, id);
            insert.setString(2, name);
            insert.setDouble(3, salary);
            insert.execute();
            System.out.println(" Employee inserted.");

            // Get salary
            CallableStatement getSal = conn.prepareCall("{CALL get_salary_by_emp_id(?, ?)}");
            getSal.setInt(1, id);
            getSal.registerOutParameter(2, Types.DECIMAL);
            getSal.execute();
            System.out.println(" Salary: " + getSal.getDouble(2));

        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

Exercise 4**4.Design a JDBC application which will demonstrate Scrollable Result Set functionality**

```
import java.sql.*;

public class Lab4
{
    public static void main(String[] args) throws ClassNotFoundException
    {
        try
        {
            // Load and register JDBC driver
            Class.forName("com.mysql.cj.jdbc.Driver");
            // create Connection
            Connection conn = DriverManager.getConnection("jdbc:mysql://localhost:3306/College","root",
"root");

            // Create statement that produces a scrollable, read-only ResultSet
            Statement stmt = conn.createStatement(ResultSet.TYPE_SCROLL_INSENSITIVE,
ResultSet.CONCUR_READ_ONLY);

            // Execute query to get employee data
            ResultSet rs = stmt.executeQuery("SELECT emp_id, name, salary FROM employee ORDER BY
emp_id");

            System.out.println("Forward iteration:");
            while (rs.next())
            { // Move forward through rows
                System.out.printf("ID: %d, Name: %s, Salary: %.2f%n",rs.getInt("emp_id"),
rs.getString("name"), rs.getDouble("salary"));
            }

            System.out.println("\nBackward iteration:");
            rs.afterLast(); // Move cursor after last row
            while (rs.previous())
            { // Move backward through rows
                System.out.printf("ID: %d, Name: %s, Salary: %.2f%n",rs.getInt("emp_id"),
rs.getString("name"), rs.getDouble("salary"));
            }

            System.out.println("\nJump to 2nd row:");
            if (rs.absolute(2))
```

```
        { // Move cursor directly to second row
            System.out.printf("ID: %d, Name: %s, Salary: %.2f%n",rs.getInt("emp_id"),
rs.getString("name"), rs.getDouble("salary"));
        }
        else
        {
            System.out.println("No second row found.");
        }
    } catch (SQLException e) {
        e.printStackTrace();
    }
}
```

Exercise 5

5.Design a JDBC application which will demonstrate Updatable Result Set functionality

```
import java.sql.*;
import java.util.Scanner;
public class Lab5 {
    public static void main(String[] args)
    {
        try
        {
            Connection con = DriverManager.getConnection("jdbc:mysql://localhost:3306/college", "root",
"root");
            Statement stmt =
con.createStatement(ResultSet.TYPE_SCROLL_SENSITIVE,ResultSet.CONCUR_UPDATABLE);
            ResultSet rs = stmt.executeQuery("SELECT emp_id, name, salary FROM employee");
            Scanner scanner = new Scanner(System.in);
            // 1. UPDATE: Increase salary of first employee by 1000
            if (rs.next()) { // move to first row
                System.out.println("Before update: " + rs.getString("name") + " - Salary: " +
rs.getDouble("salary"));

                double newSalary = rs.getDouble("salary") + 1000;
                rs.updateDouble("salary", newSalary);
                rs.updateRow();

                System.out.println("After update: " + rs.getString("name") + " - Salary: " +
rs.getDouble("salary"));
            }

            // 2. INSERT: Add a new employee row
            rs.moveToInsertRow();
            rs.updateInt("emp_id", 110);
            rs.updateString("name", "Alice");
            rs.updateDouble("salary", 55000);
            rs.insertRow();
            rs.moveToCurrentRow();
            System.out.println("Inserted new employee: Alice");

            // 3. DELETE: Delete employee by user input emp_id
            System.out.print("Enter Employee ID to delete: ");
            int deleteId = scanner.nextInt();

            boolean found = false;
            rs.beforeFirst(); // reset cursor before first row

            while (rs.next()) {
                if (rs.getInt("emp_id") == deleteId) {
                    System.out.println("Deleting employee: " + rs.getString("name"));
```

```
        rs.deleteRow();
        System.out.println("Employee deleted.");
        found = true;
        break;
    }
}

if (!found) {
    System.out.println("Employee with ID " + deleteId + " not found.");
}

} catch (SQLException e) {
    e.printStackTrace();
}
}
}
```